Anh-Minh Hoang (Eric)

23minh.ha@vinuni.edu.vn | Hanoi, Vietnam | amhoangwork-oss.github.io/erichoang/

AI alignment research focused on model representations, self-critique, and reliable safeguards for increasingly capable language models.

Current Research

Constitutional self-alignment | Work in progress

- Investigating whether self-critique and on-policy distillation can improve adherence to explicit principles with minimal supervised or RLHF intervention.

Research Manuscripts

Latent-Target Poisoning: Discovering Attack Populations in Coding Assistants

Anh Minh Hoang | 2026 | ACL ARR, October 2026 cycle · Under review

- Discovered target populations from frozen model representations and poisoned training responses to induce unwanted code on unchanged requests.
- Compared four population selectors across two model families and two datasets, measuring target emission, spillover, and coding functionality.

SafePrompt: Jailbreak Search Defense through Activation Signatures

Anh Minh Hoang | 2026 | ACL ARR, October 2026 cycle · Under review

- Developed an activation-based filter for jailbreak search that passes accepted requests unchanged to a separate language model.
- Across two models, reduced PAIR attack success from 91–92% to 14–17% at 90 queries per goal, rejecting 2.53% of 750 legitimate requests; evaluated adaptive search and sensitive-topic utility.

CSR: Physically conditioned scale-transfer super-resolution across optical missions

Minh A. Hoang and Nidal Kamel | 2026 | Under review at ISPRS Journal of Photogrammetry and Remote Sensing

- Developed satellite image reconstruction using wavelength and pixel-size information to transfer across sensors.
- Achieved the largest mean structural improvement on all four 4× OpenSR benchmark suites; evaluated transfer on 174 held-out PROBA-V scenes.

Research Experience

Center for Environmental Intelligence, VinUniversity

Research Assistant *June 2026 – Present*

- Built satellite-image training and evaluation pipelines in PyTorch Lightning and Hydra for Sentinel-2, Landsat 8/9, NAIP, and VENµS, using data accessed through Planetary Computer STAC.

Education

VinUniversity

BSc in Computer Science, in progress | Hanoi, Vietnam

Selected Honors

2025 | Regional winner, RMIT AI Hackathon; highest AU-ROC model among 400+ teams.

2025 | Top 13 among 100+ university teams, 1st Vietnamese National Collegiate AI Olympiad (OLPAI).

Skills

Research: Alignment, model representations, safety evaluation, training-data poisoning, KL-regularized adaptation.

Tools: Python, PyTorch, Transformers, LoRA/PEFT, vLLM, verl, PyTorch Lightning, Hydra.

English: IELTS 8.5 overall (Listening 9.0, Reading 9.0, Speaking 8.0, Writing 7.0).